Problem

Problem · The Fibonacci sequence and Cassini's identity. The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for every $n \geq 0$. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

1. Compute $F_2, \dots, F_6$ and A^2, A^3.
2. Prove by induction that, for every $n \geq 1$, $A^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$.
3. Prove that for all matrices M and N of order 2, $\det(MN) = \det M \times \det N$. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $N = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$.
4. Deduce that $\det(A^n) = (-1)^n$, then Cassini's identity: $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ for $n \geq 1$.
5. Check the identity for $n = 5$.
6. Using $A^{2n} = A^n \times A^n$, prove that $F_{2n} = F_n\,(F_{n+1} + F_{n-1})$ for $n \geq 1$.

Beyond the syllabus · Determinant of a product. The relation $\det(MN) = \det M \times \det N$ of question 3 is not part of the syllabus. It is proved here by computation for matrices of order 2 and is not assessed.